Pearson Edexcel International GCSE ICT (4IT1)

Paper 1: Memory & Processors Mock Examination

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Total Marks: 100 Marks

Question 1 (Total: 25 marks)

A university installs new desktop computer workstations in its main science research lab.

(a) State the definition of **Main Memory (Primary Storage)**. (1)

(b) Identify **one** form of non-volatile memory from the list below. Place a cross [X] in the correct box. (1)

[ ] A. Virtual Memory

[ ] B. Read Only Memory (ROM)

[ ] C. Random Access Memory (RAM)

[ ] D. CPU Core Cache

(c) When a user launches an application, data must be transferred between storage layers. Describe the process of loading data into **RAM** from secondary storage. (2)

(d) Explain how the total capacity of RAM installed inside a computer affects its **capacity impact** on daily user workflows. (2)

(e) When the computer systems run completely out of physical RAM workspace, the operating system triggers a fallback storage solution.

(i) State the name of this backup memory configuration space. (1)

(ii) Identify the physical hardware layer where this space is dynamically allocated. (1)

(iii) Explain **two** distinct negative performance or hardware impacts that occur when a system heavily utilizes this secondary memory storage space. (4)

Impact 1

Impact 2

(f) Modern computing devices require permanent data blocks to initiate hardware check routines.

(i) State what is meant by **Read Only Memory (ROM)**. (1)

(ii) State the primary function of a ROM chip inside a **single-purpose device** compared to a **general-purpose computer**. (2)

Single-Purpose

General-Purpose

(g) Complete the following comparison table using the information from the textbook guidelines. (4)

Feature	Random Access Memory (RAM)	Read Only Memory (ROM)
Saves data when power is cut?		
Can the user physically upgrade it?		

(h) State **two** distinct core modifications or instruction rules that differentiate RAM from ROM. (2)

1.

2.

Question 2 (Total: 25 marks)

Specialist electronic engineers categorize hardware memory chips into distinct classifications based on how data blocks can be written or erased.

(a) Define the functional attributes of the following **four** specialized types of ROM: (8)

1. Mask Programmed ROM:

2. PROM (Programmable ROM):

3. EPROM (Erasable Programmable ROM):

4. EEPROM (Electrically Erasable Programmable ROM):

(b) Explain how **EEPROM** parameters allow manufacturers to update system firmware configurations without physically removing the chip from the motherboard circuit boards. (3)

(c) Flash memory is described as a highly advanced derivative architecture type of EEPROM.

(i) State **three key physical features** of Flash memory configurations. (3)

1.

2.

3.

(ii) Identify **three common technological deployments** that use Flash memory as their core storage technology. (3)

4.

5.

6.

(d) Explain **two reasons** why the technical characteristics of flash memory configurations make it uniquely suitable for use as storage inside portable smartphones. (4)

Reason 1

Reason 2

(e) Differentiate between how **volatile architectures** manage data pathways compared to **non-volatile storage systems** when electrical energy inputs are completely removed. (4)

Question 3 (Total: 25 marks)

Processor architecture and operational execution loops dictate how fast instructions are computed.

(a) Define the three stages involved inside **The Processor Cycle** (CPU execution loop). (3)

Stage 1

Stage 2

Stage 3

(b) Explain what is meant by a **CPU core**. (2)

(c) A network specialist configures a system utilizing a quad-core processor setup. Describe how a **quad-core processor** processes work instructions differently than a legacy single-core processing chip. (2)

(d) Multi-core architectures alter system energy outputs and performance metrics. Explain **three** energy and mechanical advantages gained by deploying multi-core processing configurations inside portable computer builds. (6)

Advantage 1

Advantage 2

Advantage 3

(e) Processor execution throughput parameters are evaluated using specific system clock benchmarks. State the exact technical frequency value in cycles per second represented by each metric below: (4)

- Hertz (Hz):

- Kilohertz (kHz):

- Megahertz (MHz):

- Gigahertz (GHz):

(f) Identify **one** statement that correctly describes a processor's clock cycle. Place a cross [X] in the correct box. (1)

[   ] A. The number of times per second that a processor can carry out instructions.

[   ] B. The number of instructions that a processor can carry out each second.

[   ]] C. The number of times that a processor can load instructions for processing.

[   ] D. The number of times that each process repeats in a processor.

(g) An advertisement for a computer system claims that a 3.5 GHz CPU will always perform calculations faster than a 2.8 GHz CPU. Explain why this statement is considered an **Exam Trap** by analyzing how differing processor engineering layouts process instructions. (3)

(h) State what actions must occur to any software file stored inside a computer's secondary storage before it can be actively computed by the processor. (2)

Question 4: Structural Architecture Analysis (Total: 25 marks)

A corporate office space plans to swap its fleet of old terminal systems for modern energy-efficient portable setups.

Evaluate the hardware design choices that the network consultants must make when deciding between systems featuring standard upgradable high-capacity mechanical RAM structures versus devices that rely entirely on low-power solid-state Flash-based components.

In your evaluation, you must discuss the direct impacts on **system latency** (clock throughput and cycle loading), **power efficiency footprints**, **hardware component longevity**, and **total structural upgrade parameters**. (25)

Plan your answer carefully. Marks will be awarded for structured technical progression, correct usage of ICT terms from the specification text, balanced comparative metrics, and a logical conclusion.

Structured Essay Response Area